UE24CS352A Machine Learning Mini-Project — Problem Statement 6

Pancreatic Cancer Prognosis Using Clinical and Radiomic Data

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1. Problem Statement

Pancreatic ductal adenocarcinoma is rarely detected at early resectable stages; most patients present with borderline resectable or locally advanced tumors where accurate survival prognosis is critical for personalizing chemotherapy timing and radiation regimens. Replicating the machine learning formulation of Arash Jamalian (Stanford CS229, 2020), this project develops a multimodal prognostic pipeline integrating 12 clinical variables with 900 quantitative 3D MRI radiomic features. Core methodological challenges include: (a) small sample size (n = 80), (b) right-censored survival times, (c) severe minority class imbalance in the high-risk (< 1 year) regime, and (d) high-dimensional multicollinearity across 900 radiomic texture, shape, and wavelet descriptors.

2. Dataset Details (Synthetic Cohort)

Dataset Origin & Disclaimer: The original Stanford Cancer Center QFIP dataset is protected under NDA and cannot be redistributed. All experiments utilize a **synthetic cohort of 80 patients** generated via src/data_generator.py (seed = 42) matching the structural dimensionality of the reference study. These records are **NOT real patient data**, and metrics must not be read as outperforming the paper.

• **Clinical Variables (12):** Stage Grouping (IB, IIA, IIB, III), Chemotherapy Time (Unchecked vs. Post-SBRT), Radiation Therapy (SBRT vs. Conventional), NCCN Resectability, Tumor Location, Primary Tumor Stage (T1–T4), Karnofsky Performance Status (KPS: 60–100), Gender, and 3D tumor dimensions/volume (Width, Height, Depth, Volume).

• **Radiomic Feature Space (900):** PyRadiomics/QFIP descriptors across first-order intensity, 2D/3D shape, and texture matrices (GLCM, GLSZM, GLRLM) filtered through 3D wavelets (LLL–HHH) and Laplacian-of-Gaussian (LoG 1/3/5 mm). Collinear redundancy was simulated via 10 shared latent components.

• **Survival & Censoring Simulation:** High-risk times simulated as $U(135, 266)$ days; low-risk as Gamma(2.4, 145) + 272 days. Right-censoring affected 17.5% of cases (14/80) sampled among patients surviving > 310 days.

• **SMOTE Augmentation:** Expanded the cohort from 80 to **112 patients (+32 synthetic high-risk cases)**.

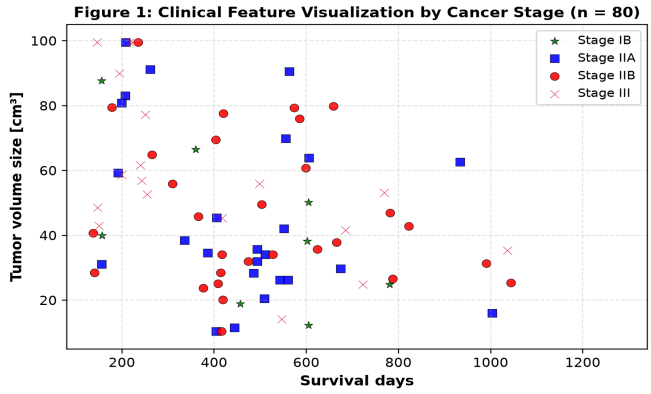
3. Approach & Mathematical Formulation

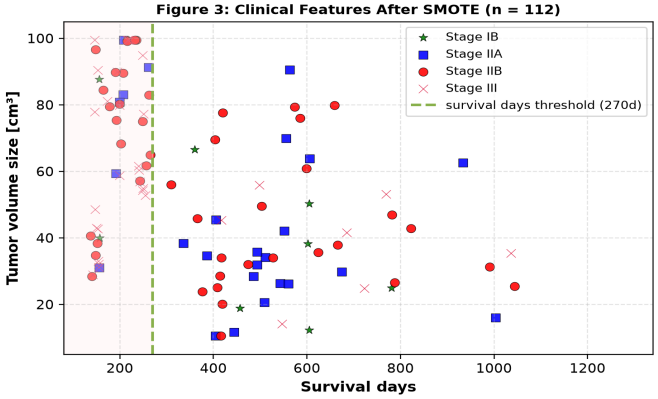
(A) Naive Bayes Threshold Selection: Five clinical features are discretized into V bins and evaluated across candidate survival boundaries (240, 270, 300, 330, 360 days) via 10-fold CV using Laplace-smoothed Multinomial Naive Bayes: $\phi_{dk|y=c} = (1 + \sum 1\{x_d=k, y=c\})/(V_d + \sum 1\{y=c\})$. Thresholds 270, 300, and 330 days tie at 0.74 balanced accuracy. The **270-day boundary** is chosen deterministically via the explicit tie-breaking rule: lowest fold standard deviation (0.105 for 270d/300d vs. 0.142 for 330d), then smallest threshold (270d < 300d).

(B) Leak-Free Cross-Validation Pipeline: To eliminate severe data leakage present in the paper-style pipeline (where SMOTE and pruning occurred globally before CV), we isolate all preprocessing inside each CV fold: (1) SMOTE (k=5) is applied exclusively to the training split, (2) 410 collinear radiomic features with error bands > 1.0 are pruned using training split statistics only, and (3) validation folds contain strictly original patients.

(C) L2-Regularized Cox Proportional Hazards Regression: Semi-parametric partial likelihood maximization models hazard as $h(t, X) = h_0(t) \exp(\beta^T X)$ with an L2 Ridge penalty: $0.5 \times penalizer \times ||\beta||_2^2$ (penalizer = 0.1, l1_ratio = 0.0). No L1 or ElasticNet penalty is applied. Risk discrimination is quantified via Harrell's Concordance Index (C-index).

(D) Kaplan-Meier & Log-Rank Hypothesis Testing: Patients are stratified into high- and low-risk groups at the median partial hazard score, comparing survival trajectories via non-parametric Kaplan-Meier estimators and the log-rank test statistic.





4. Implementation Overview & Experimental Results

The system is implemented in Python across modular packages: src/data_generator.py, src/naive_bayes_smote.py, src/survival_models.py, and src/visualization.py. Interactive demonstration is provided via app.py (Streamlit) and demo_cli.py (Terminal).

Table 1: Naive Bayes 10-Fold CV Balanced Accuracy across Survival Thresholds

Threshold	High / Low	10-Fold Bal. Acc.	Paper (Context)	Selection Decision
240 days	18 / 62	0.64 ± 0.172	0.53	Suboptimal
270 days	24 / 56	**0.74 ± 0.105**	0.69	Optimal (Tied at 0.74; lowest std 0.105, smallest threshold 270d)
300 days	24 / 56	0.74 ± 0.105	0.61	Tied at 0.74 (std=0.105; discarded in favor of smaller 270d cutoff)
330 days	25 / 55	0.74 ± 0.142	0.57	Tied at 0.74 (std=0.142; discarded due to higher fold variance)
360 days	26 / 54	0.72 ± 0.166	0.61	Suboptimal

Table 2 & 2b: 10-Fold CV Concordance Index (Leak-Free vs. Leaky Comparison)

Model Feature Set	# Features	Leak-Free C-Index	Leaky C-Index	Leakage Delta	Paper (Real Data Context)
Clinical Features	12	**0.71 ± 0.088**	0.74 ± 0.101	+0.03	0.67
Radiomic Features	490	**0.77 ± 0.114**	0.83 ± 0.083	+0.06	0.76
Clinical and Radiomic Features	502	**0.80 ± 0.112**	0.86 ± 0.066	+0.06	0.82

Figure 4: KM Survival Curves (Log-Rank=135.1, p < 0.001)

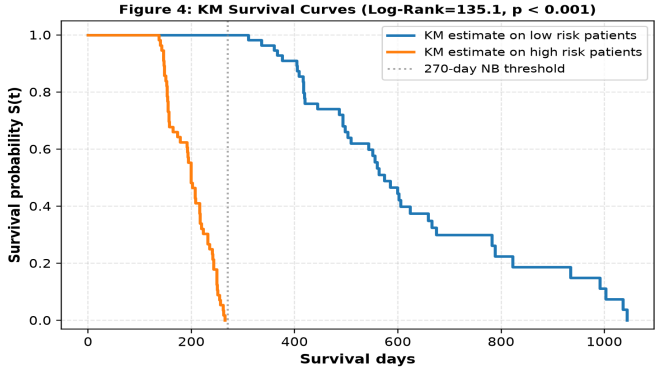
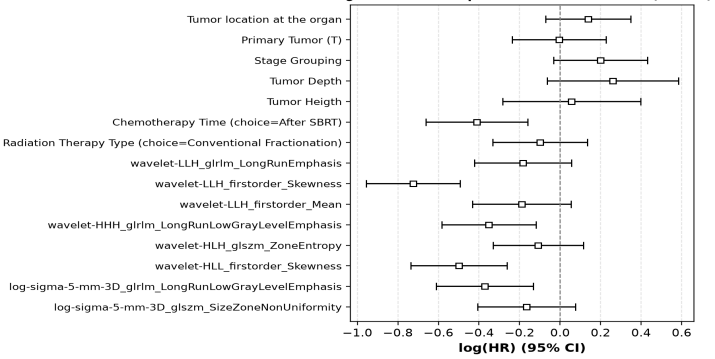


Figure 5: CoxPH Top 15 Feature Coefficients (95% CI)



5. Conclusions & Honest Scientific Assessment

• **Leak-Free Findings:** Honest leak-free cross-validation yielded C-indices of **0.71 ± 0.09** (Clinical), **0.77 ± 0.11** (Radiomic), and **0.80 ± 0.11** (Combined). Multimodal fusion consistently outperformed single-modality baselines (+0.09 over Clinical).

• **Data Leakage Impact:** Performing SMOTE and pruning before CV artificially inflated C-indices by **+0.03 to +0.06** (to 0.74, 0.83, 0.86). Only the leak-free numbers reflect genuine generalizable performance.

• **Risk Stratification:** Median hazard splitting achieved marked survival separation (Log-Rank statistic = **135.11**, $p < 0.001$). Chemotherapy post-SBRT was protective (log HR < 0), while advanced primary tumor stage (T2–T4) accelerated hazard.

• **Limitations:** (1) Results are derived from synthetic data and cannot be compared to clinical outcomes in the paper. (2) Small cohort size ($n = 80$) induces substantial cross-validation variance. (3) Euclidean SMOTE does not account for survival censoring mechanisms. (4) Proportional hazards assumptions were not formally tested via Schoenfeld residuals.